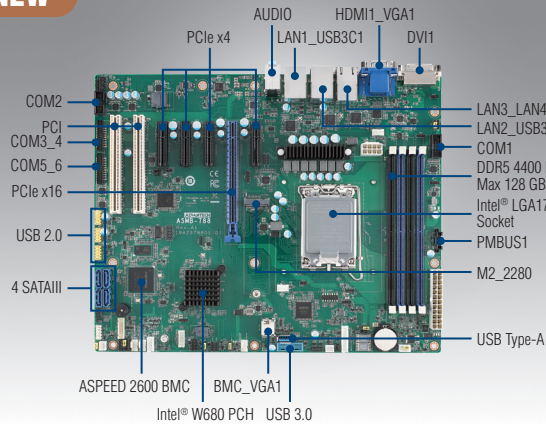


ASMB-788

LGA 1700 Intel® 14th/13th/ 12th Gen. Core™ ATX Server Board with 4 x DDR5, 5 x PCIe (1 x Gen 5, 3 x Gen 4, 1x Gen 3), 7 x USB 3.2, 4 x SATA3, Quad/Dual LANs, and IPMI

NEW



Features

- LGA 1700 Intel® 14th/13th/ 12th Gen. Core™ i9/i7/i5/i3 processors with W680 chipset
- DDR5 ECC/Non-ECC 5600/4400/4000/3600 MT/s UDIMM up to 128 GB
- One PCIe x16 (Gen 5) and four PCIe x4 (3 x Gen 4 & 1 x Gen 3) slots
- Triple displays - DVI-D, VGA, and HDMI 2.0 ports
- Four SATA 3 ports and seven USB 3.2 ports
- One M.2 2280 (PCIe x4)
- Rackmount optimized placement with positive air flow design
- 0 ~ 60 °C (32 ~ 140 °F) ambient operating temperature range

Note: Supported OS including Windows 10/ 11 (64-bit) and Server 2022 (64-bit) have been validated.

CE FCC

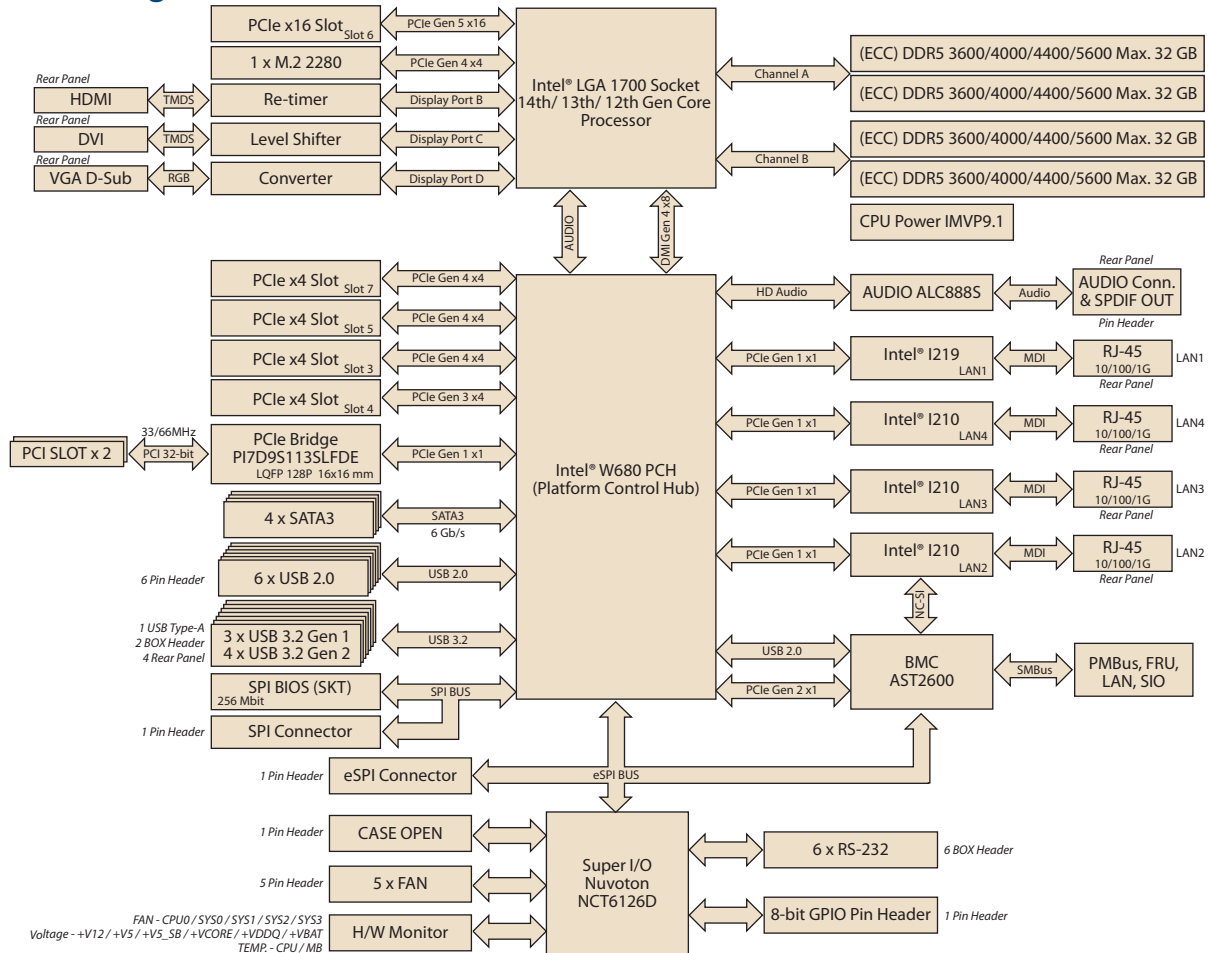
Specifications

Form Factor		ATX form factor				
Processor System	Socket	Intel® LGA 1700				
	CPU type	14th/13th/ 12th Gen Intel® Core™ i9/i7/i5/i3 series processors (TDP up to 125W)				
Core Logic	Chipset	Intel® W680 chipset				
Memory	Total slots	4 x (Dual Channel)				
	Max. capacity	128 GB				
	Memory type	DDR5 ECC*/Non-ECC 5600/4400**/4000/3600 MT/s UDIMM (288-pin)				
	Memory size	8 GB, 16 GB, 32 GB				
Expansion Slots	Total PCI/PCIe slots	7				
	Slot location 1	PCI slot				
	Slot location 2	PCI slot				
	Slot location 3	PCIe x4 slot (Gen 4 x4 link)				
	Slot location 4	PCIe x4 slot (Gen 3 x4 link)				
	Slot location 5	PCIe x4 slot (Gen 4 x4 link)				
	Slot location 6	PCIe x16 slot (Gen 5 x16 link)				
	Slot location 7	PCIe x4 slot (Gen 4 x4 link)				
Graphics	Controller	Intel® HD Graphics				
	VRAM	1 GB maximum shared memory with 2 GB and above system memory installed				
Storage	SATA controller	Intel® W680: 4 SATA3 6 Gb/s ports with Intel Rapid Storage Technology support (for Windows only) (Supports software RAID 0, 1, 10, & 5)				
	SAS controller	-				
Networking	Interface	10/100/1000 Mbps				
	Controller	GbE LAN1: Intel® I219; GbE LAN 2~4: Intel® I210-AT				
	Connector	RJ-45 x 2 for G2 version; RJ-45 x 4 for G4 version				
Onboard I/O Connector	Printer Port connector	-				
	PSU connector	24-pin SSI power connector + 8-pin SSI 12V power connector				
	USB connector	9, 1 x USB 3.2 Gen 1 (2 ports), 3 x USB 2.0 (6 ports), 1 x USB 3.2 Gen 1 Type-A				
	M.2 connector	1 x M.2 2280 (PCIe)				
	Fan header	5 x (4-pin)				
	SMBus	1				
	Chassis intrusion	1				
	Front LAN LED	2				
	Serial port header	5, 2 x RS-232 dual-port (4 ports), 1 x RS-232 single-port				
	Audio connector	1 x (ALC 888S)				
Rear I/O Connector	PS/2 KB/Mouse	-				
	External serial port	1 (via COM cable in the package)				
	External USB port	4 x (USB 3.2 Gen 2)				
	VGA port	1 x (Resolution up to 1920 x 1200 @ 60 Hz)				
	DVI port	1 x (Resolution up to 1920 x 1200 @ 60 Hz)				
	HDMI port	1 x (Resolution up to 4096 x 2160 @ 60 Hz)				
	RJ-45	2 for G2 version; 4 for G4 version				
	Audio connector	2 x (Mic-in, Line-out)				
Watchdog Timer	PS/2 KB/Mouse	-				
	Output	System reset				
Power Requirement	Interval	Programmable 1 ~ 255 sec				
	Power On	3.3 V	5 V	12 V	12 V_8P	5 V _{SB}
Environment	(Core i5-12500 3.0GHz DDR5 4800 16 GB x 4)	2.05 A	1.69 A	0.36 A	13 A	0.10 A
	Operating	Non-operating				
	Temperature	0 ~ 60 °C, depends on CPU speed and cooler solution				
Physical Characteristics	Humidity	-40 ~ 85 °C				
		40 °C @ 95% RH Non-condensing				
	Dimensions	60 °C @ 95% RH Non-condensing				
	Dimensions	304.8 x 244 mm (12 x 9.6 in)				

* ECC will not be supported when paired with Core i3 processors.

** 4400MT/s can only be supported with 1 DIMM per channel configuration (2 DIMMs in total).

Block Diagram



Ordering Information

Part Number	Chipset	Memory	Display	LAN	SATAIII	IPMI
ASMB-788G2-00A1	W680	DDR5 Non-ECC/ECC UDIMM	VGA + DVI-D + HDMI	2	4	-
ASMB-788G4-00A1	W680	DDR5 Non-ECC/ECC UDIMM	VGA + DVI-D + HDMI	4	4	Y
ASMB-788G4-00A1U	W680	DDR5 Non-ECC/ECC UDIMM	VGA + DVI-D + HDMI	4	4	Y

Packing List

Part Number	Description	Quantity
2042078800	Startup manual	x 1
1700003194	SATA data cables	x 2
1700022749-11	SATA power cables	x 2
1700001788	COM1 cable	x 1
1960088021T011	I/O shield	x 1
1930005673-11	Screw for M.2 device	x 1

Optional Accessories

Part Number	Description
1700029038-01	BMC VGA cable with bracket (35 cm / 13.7 in, for G4 SKU)

Note: If you need BMC KVM function, please select this cable.

Part Number	Description
1970005958N001	LGA 1700 CPU Cooler for 3U/4U Tower Cooler (CPU 230W at ambient 30°C)
1970005436T001	LGA 1700 CPU cooler for 2U/4U chassis (CPU 125W)
1970005349T000	LGA 1700 CPU cooler for 2U/4U chassis (CPU under 65W)
1700002204	Dual-port USB 2.0 cable (27 cm / 10.6 in) with bracket
17000008461	Four-port USB 2.0 cable (30.5 cm / 12 in) with bracket
1700028292-01	Dual-port USB 3.0 cable (30 cm / 11.8 in) with bracket
1701090401	Single port RS-232 COM cable (40 cm / 15.7 in) with bracket
1700026254-01	Dual-port RS-232 COM cable (45 cm / 17.7 in) with bracket
PCA-TPMSPI-00A1	TPM 2.0 Module by SPI

Note: Users need to purchase the ASMB-788 proprietary CPU cooler from Advantech. Other brands of CPU cooler may NOT be compatible with ASMB-788.

I/O View



ASMB-788G4-00A1



ASMB-788G2-00A1

ACP-2010/2320

2U Rackmount Chassis for ATX/ MicroATX Motherboard with Low-Profile Rear Bracket Option



Features

- LED indicators and audible alarm notification for system fault detection
- ACP-2010: One 5.25" and three 3.5 shock-resistant drive bays
- ACP-2320: One Slim optical disk drive, two internal 3.5" drives and dual hot-swap SATA HDD trays
- Various riser card options to support three full-size PCI or PCIe cards for expansion
- Supports 250W /350W single power supply
- Reusable and washable air filters
- Built-in Intelligent System Module enabling whole system fan control and remote manageability

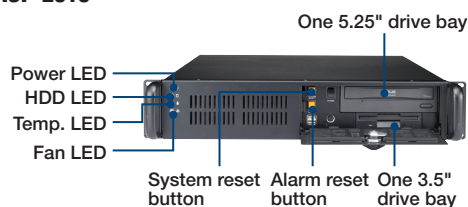
CE FCC

Specifications

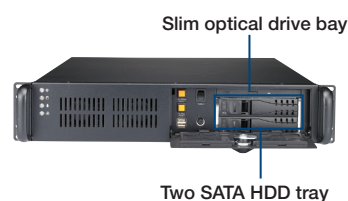
Drive Bay		External	Internal
	Slim Optical Disk Drive	- / 1	-
	5.25"	1 / 0	-
	3.5"	1 / 2	2 / 2
	Hot Swap	- / 2 (Compatible with both 3.5" or 2.5" devices, no convert bracket needed)	
Cooling	Fan	2 (8 cm PWM /47 CFM each) / 2 (8 cm PWM /47 CFM each) + 1 (6 cm PWM /28 CFM) for SATA storage unit	
	Air Filters	Yes	
	Smart Fan	System fan speed referring to chassis thermal sensor, or CPU temperature (Available with selected Advantech MBs/CPU cards)	
Front I/O Interface	USB 2.0	2	
	PS/2	1	
Miscellaneous	LED Indicators	Power, HDD, temperature, fan	
	Rear Panel	Two 9-pin D-Sub openings	
Intelligent System Module Hardware Monitoring	CPU/system fan speed, CPU/system temperature (Only Advantech AIMB-78X, PCE-51XX/71XX mainboards can enable CPU fan speed and temperature monitoring function of Intelligent System Module)		
Environment		Operating	Non-operating
	Temperature	0 ~ 40 °C (32 ~ 104 °F)	-20 ~ 60 °C (-4 ~ 140 °F)
	Humidity	10 ~ 85% @ 40 °C, non-condensing	10 ~ 95% @ 40 °C, non-condensing
	Vibration (5 ~ 500 Hz)	1 Grms	2 G
Physical Characteristics	Shock	10 G (with 11 ms duration, half sine wave)	
	Dimensions (W x H x D)	482 x 88 x 480 mm (19" x 3.46" x 18.9")	
	Weight	10.7 kg (23.5 lb) / 11.7 kg (25.7 lb)	

Front View

ACP-2010



ACP-2320



Rear View

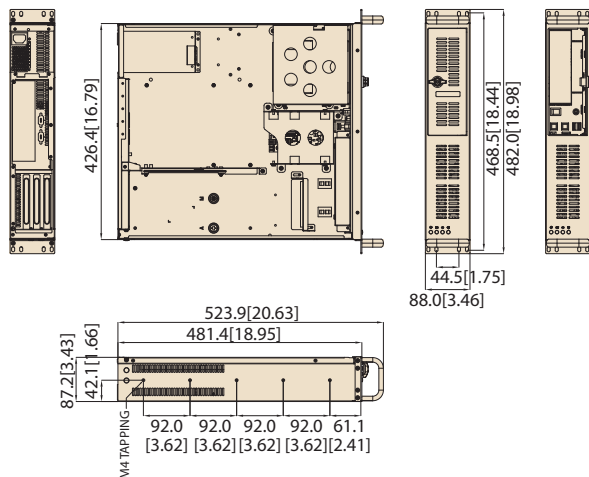


Rear I/O bracket for low profile add-on cards

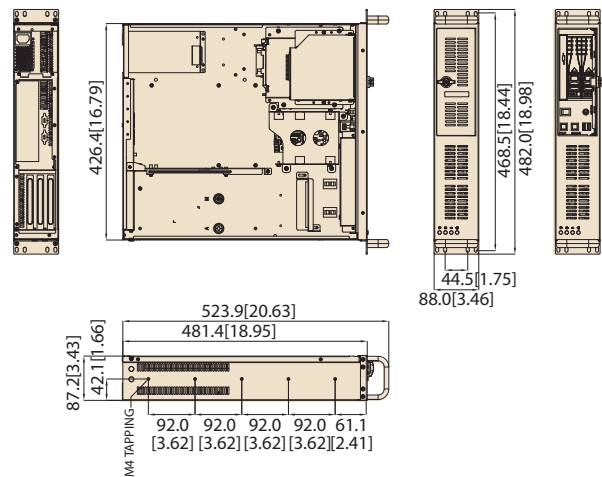
Dimensions

Unit: mm [inch]

ACP-2010



ACP-2320



Ordering Information

Part Number	Power supply	Power supply type	Wattage	80 plus grade	Switch type
ACP-2010MB-25D	PS8-250FATX-BB	Single	250W	Bronze	ATX
ACP-2010MB-35D	PS8-350FATX-GB	Single	350W	Gold	ATX
ACP-2320MB-35D	PS8-350FATX-GB	Single	350W	Gold	ATX

Packing List

Part number	Description	Quantity
ACP-2010 / ACP-2320, Rev.D	2U rackmount chassis with handle set	1
1700018888	chassis power on/off switch cable	1
1750009329-01	System fan - 8cm	2
1700030260-01	system fan cable	2
1750009289-01	hot-swap module fan - 6cm (in ACP-2320)	1
1700024778-11	hot-swap module fan cable (in ACP-2320)	1
1700026158-01	4P Y cable for system fan (in ACP-2320)	1
1990007986S000	Filter behind the door 156 x 45 x 5 mm	1
1990001949	Filter in the front of the fans 168 x 58 mm	1
98R2132000E	hot-swap tray HDD drawer (compatible with both 2.5" and 3.5" HDD/SSD)	2
Content of Accessory Box		
98R22010000	Handle set	1
	Screws/nuts and rubber for motherboard and devices	one bag
2001201000	User manual	1

Optional Accessories

Part Number	Description	
96CB-SATAPOWER-6P2	SATA power wire 6-pin to big 4-pin (for slim ODD)	
IPC-DT-3120E	Mobile rack for converting a 3.5" drive bay to dual 2.5" SATA HDD trays	
Rear I/O panel kit for low-profile add-on cards		
Part number	Description	Quantity
1939000361	Screws	7
1960074850T000	I/O bracket	7
1960019475T206	2U Rear panel for low-profile	1